

Process Scheduling Algorithms

- First Come First Served runs jobs in the order they arrive, which is simple but can cause the convoy effect when a long job blocks short ones.
- Round Robin gives each process a fixed time quantum.
- Complexity is $O(n \log n)$ with a priority queue.

Comparison

Preemptive scheduling can interrupt a running process when a higher priority process becomes ready to run.

Non-preemptive scheduling waits until the running process finishes or blocks before choosing the next one.

Algorithm	Average waiting time	Starvation
FCFS	High when jobs vary	No
SJF	Lowest possible	Yes